

Uzair Ahmed

AI/ML Researcher

Daejeon, Korea · uziahmd@outlook.com · +82-10-6541-1277
GitHub: github.com/uziahmd · LinkedIn: linkedin.com/in/uziahmd

Professional Summary

AI researcher building production vision-language and LLM systems backed by rigorous evaluation. At **KPST**, shipped an infinite-stream industrial-safety VLM (Qwen2.5-VL-7B + StreamingVLM) that monitors never-ending camera feeds with **bounded memory** and emits schema-constrained hazard decisions for a live alarm system. At KAIST's **Knowledge & Democracy Lab**, research LLM reasoning and difficulty perception, VLM bias and interpretability, and LLM safety (adversarial prefills, many-shot jailbreaking, introspection). Co-authored an **IMWUT/Ubicomp 2024** reproducible mobile-sensing pipeline and led its personalization component (up to ~17% AUROC gain). Strengths: PyTorch, Hugging Face (Transformers, PEFT, TRL), LLM/VLM fine-tuning (LoRA/QLoRA), interpretability and evaluation-harness design, and reproducible experimentation.

Education

Korea Advanced Institute of Science & Technology (KAIST) Sep 2020 – Feb 2026
B.S. in Computer Science & Business Technology Management · GPA: 3.3/4.3

Experience

KPST (Korea Platform Service Technology) — AI Researcher Feb 2026 – Present
Infinite-Memory Streaming VLM for Industrial Hazard Detection

- Shipped a real-time industrial-safety VLM that watches a continuous, never-ending camera stream and emits one compact, schema-constrained JSON hazard decision per step (~2 Hz) to drive a downstream alarm system — handling two tasks (**forklift entry hazard**, **robot-zone intrusion**) via prompt switching, end-to-end with no separate object detector or bounding-box head.
- Achieved **constant memory and per-step latency regardless of stream duration** by adapting **Qwen2.5-VL-7B** with **StreamingVLM**-style streaming attention (attention-sink + sliding-window KV cache, contiguous 3D RoPE), keeping steady-state KV at a few thousand tokens so a camera can be monitored for hours or days without KV growth or drift.
- Designed a **two-stage LoRA SFT** pipeline (Stage 1: broad supervised fine-tuning over fixed windows; Stage 2: event-centered annealing on hazard transitions) on a **single NVIDIA V100 32 GB** (fp16, SDPA), freezing the vision tower to keep trainable parameters a small fraction of the 7B base.
- Guaranteed **100% output validity** via FSM / schema-constrained decoding over a fixed compact JSON schema (hazard label/present, zone relation, object state/direction), enabling reliable downstream consumption instead of free-form text.
- Engineered the data and evaluation pipeline: built streaming training trees from raw CCTV and per-frame labels aligned by true probed frame index, applied conservative safety-biased label aggregation, and measured generalization on **held-out cameras never seen in training** (not random clips).
- Stack*: Python; PyTorch; Hugging Face Transformers; StreamingVLM; PEFT (LoRA); CUDA; FFmpeg; evaluation harnesses.

Knowledge & Democracy Lab — KAIST | Undergraduate Researcher Aug 2025 – Feb 2026

- LLM Difficulty Perception (LLM Reasoning)** — Built from scratch a sandboxed code-execution framework with strict I/O judging for the competitive-programming track (E2H-Codeforces, 200 problems) and contributed to a reasoning-evaluation suite spanning **4 benchmarks** (AIME, E2H-Codeforces, E2H-ARC, LLM-Coordination; 658 problems) across **6 reasoning LLMs** to study how difficulty framing affects performance; implemented the **LoRA/QLoRA** fine-tuning pipeline (TRL + PEFT) and curated AIME / E2H-Codeforces SFT data (chat-format conversion, multi-solution support, dedup). (*submitted ACL 2026*)
- LLM Safety & Introspection** — Built a reproducible pipeline (SGLang + OpenRouter) evaluating pre-fill attacks (control / static / adversarial) and model introspection over **HarmBench** and **SocialHarmBench**; implemented **ADVPrefix** end-to-end (candidate generation, ablation, NLL scoring, PASR eval-

uation, optimized-prefix selection) and reproduced Anthropic’s **Many-Shot Jailbreaking** (harmful-demo dataset construction, Llama Guard scoring, attack-vs-shot-count analysis) across Gemma/Qwen/Llama.

- **VLM Ideology & Interpretability** — Extended ideology-probing work to Vision-Language Models: trained per-layer/per-head ridge-regression **linear probes** on **DW-NOMINATE** scores, performed inference-time steering by injecting learned ideology vectors, built a ~550-politician multimodal dataset (Voteview + portraits), and ran **NMF topic modeling** on chain-of-thought to surface visual cues models associate with ideology.
- **Bias Localization in VLMs** — Probing sensitive-attribute leakage and spurious correlations in **InternVL**, **Gemma3-VL**, **Qwen-VL** via layer-wise linear probes and attribution (Grad-CAM, Integrated Gradients, TCAV); applying counterfactual edits to quantify subgroup gaps and localize bias to image regions and cross-modal tokens. (*ongoing*)

ICLab — KAIST | Undergraduate Researcher

May 2023 – Jun 2025

- Co-authored an **IMWUT/Ubicomp 2024** reproducible in-the-wild stress-prediction pipeline (feature extraction, leakage guards, temporal-order controls); led the personalization component (partial / multi-task personalization), improving **AUROC by up to ~17%** on the DeepStress dataset over the user-independent baseline.
- Built **CrossShift**, a framework quantifying user-level data shift across **8 datasets / 693 users**: PERMANOVA/PERMDISP and χ^2 testing with Wasserstein (covariate), L2 (label), OTDD (conditional), and JS-divergence (concept) quantification, a LOSO harness, and domain-generalization baselines — finding **concept shift** $P(Y|X)$ the primary driver of cross-user degradation ($\beta = 0.265$) and using JS-based OOD checks for onboarding-time selective personalization. (*under review*)
- Integrated multimodal sensor data (Audio, Aqara, Withings, Smartphone) with KNN imputation and feature selection (ANOVA, LASSO-CV, correlation filtering); identified 60-minute windows and “Yesterday Epoch” features as the strongest configuration via LOSO and K-Fold CV with HPO.
- Designed the **CausalCFF** pipeline (POI clustering, association-rule mining, Convergent Cross Mapping causal analysis) for a **CHI EA 2025** extended abstract; prepared the codebase for public release.

DocsHunt AI | AI Engineer (Seoul, South Korea)

Jun 2024 – Aug 2024

- Designed and shipped an automated prompt-template pipeline for multilingual document automation (Python, Flask, LangChain), reducing prompt variance and failure modes across languages and document types.
- Productionized services on Naver Cloud Platform: hardened REST APIs with pytest-based tests, structured logging, and retry-safe handlers; profiled and tuned Flask + Gunicorn (workers, concurrency, timeouts) to roughly triple sustained throughput and stabilize p95/p99 tail latency under load.

Publications & Preprints

- P. Zhang, G. Jung, J. Alikhanov, **U. Ahmed**, U. Lee. *A Reproducible Stress Prediction Pipeline with Mobile Sensor Data*. Proc. ACM IMWUT 8(3), Art. 143, Aug 2024 (35pp).
- P. Zhang, G. Jung, **U. Ahmed**, U. Lee. *CausalCFF: Causal Analysis between User Stress Level and Contextually Filtered Features*. CHI EA '25, 2025.
- **U. Ahmed**, M. F. Zahid, H. Ahmad, A. Oh, J. Thorne. *Standing on the Shoulders of Language Giants: Improving LLM Capabilities on Low-Resource Languages through Pivots*. JKAIA, 2023.
- Q. M. Nguyen, **U. Ahmed**, T. Kim. *Difficulty Perception in the Reasoning of LLMs*. Under review (ACL 2026).
- P. Zhang, M. Park, T. Ismatzoda, A. Mustafakulov, **U. Ahmed**, O. Najimov, J. Alikhanov, S. Ghosh, U. Lee. *CrossShift: Quantifying Inter-Personal Differences in Mobile Sensing for Mental Health*. Under review.

Selected Projects

Smart Binarization | Activation-Aware Selective Binarization for LLM Quantization

- Built a selective-binarization PTQ framework for LLMs (GPTQ-based), supporting binary/sign and low-bit variants with pluggable saliency methods.

- Implemented **activation-aware saliency**: collected activation energy during calibration to prioritize weights that matter most under real inference (vs. magnitude-only), with per-layer importance, global budget allocation, top- K masking, and mixed-precision layers that keep salient weights high-precision while binarizing the rest.
- *Results (WikiText-2 PPL, 50% salient)*: kept perplexity within $\sim 5\%$ of full precision on OPT-1.3B (15.42 vs 14.62), while reducing the perplexity gap versus magnitude-only selection by up to $64\times$ (Gemma-3-1B), $18\times$ (OPT-125M), and $3.3\times$ (OPT-1.3B).
- *Stack*: Python; PyTorch; Hugging Face Transformers; GPTQ/PTQ; CUDA.

Allah-u-Allam | Hierarchical RAG

2025 · NDA · Freelance

- Developed a hierarchical Retrieval-Augmented Generation system over **96+** canonical texts enabling context-aware, multi-level reasoning; fine-tuned a **70B**-class LLM with PEFT (LoRA, QLoRA) for secure, offline (air-gapped) inference via **Ollama**.
- Designed a modular retrieval layer with **FAISS** (HNSW/IVF-PQ), semantic chunking, and taxonomy-aware indexing; orchestrated agentic planning, tool use, and answer verification with **LangChain/LangGraph**, containerized with **Docker** for local deployment.

Pivot-Assisted Consensus (PAC) for Low-Resource Languages | KAIST

2023 · First Author

- First-authored a **zero-shot** approach (no fine-tuning) that routes low-resource-language inputs through multiple high-resource pivots (e.g., English/Spanish) and aggregates the model judgments into a **consensus** prediction; evaluated across **MGSM**, **XQuAD**, **XNLI** (plus Urdu sentiment and QA), improving translation METEOR by up to $+16.8\%$ (Hindi) and reasoning exact-match by $+15.1\%$ over single-pass baselines.

Signature Forgery Verification on ChiSig | Vision Transformers and CNNs

2024

- Built a data pipeline over 10,242 signatures (500 identities) and produced embeddings (ResNet50, Inception-ResNet, ResNeXt50, VGG16, ViT) with cosine-similarity pair verification and tuned decision thresholds.
- *Results*: Inception-ResNet 85.29% accuracy (best); ViT 81.26%; reported EER and TAR at FAR = 0.1%.

Awards & Activities

- KAIST Lean Start-up Industry Cooperation — **Winner**
- National Mathematics Olympiad — **Top 5 in Pakistan (2018, 2019)**
- Google Code-in — **Runner-up (2018)**
- WorldQuant Challenge — **Gold**; International Quant Competition — **Top 20%**
- Cambridge IGCSE Distinction (Pakistan Top) in International Mathematics (2019); (Region Top) in Mathematics (2018)

Technical Skills

Programming: Python; C; OCaml; Kotlin; SQL; Bash

ML/DL: PyTorch; TensorFlow; scikit-learn; XGBoost; Hugging Face (Transformers, PEFT, TRL); OpenCLIP; LangChain/LangGraph

LLM/VLM: LoRA/QLoRA fine-tuning; PTQ/quantization; RAG; FAISS; Ollama; SGLang; prompt/eval design; constrained decoding; interpretability (linear probes, Grad-CAM, Integrated Gradients, TCAV, CKA)

Tooling & Infra: Hydra; Weights & Biases; Docker; Git; pytest; Flask; Unicorn; REST APIs; Linux; CUDA; OpenCV

Cloud: Naver Cloud Platform; AWS (S3, EC2)

Languages

English (fluent); Urdu (native); Korean (Beginner — TOPIK 2)

Coursework

Data Structures; Algorithms; Operating Systems (+ Lab); Systems Programming; Programming Languages; Databases; Discrete Math; Linear Algebra; Machine Learning; NLP; ML for Computer Vision; Intro to Information Security; Generative AI for Society; Technology/Innovation/Financial Management.